Problem 3

Problem 3

Problem 1 *Let $f(x) = 2x^3 - 4x^2 + 1151$ and $g(x) = x^3 \sin(x)$. Compute $f^{(4)}(x)$ and $g^{(4)}(x)$.*

Explanation. Let's find the first four derivatives of f.

$$
\begin{aligned}
f^{(0)}(x) &= 2x^3 - 4x^2 + 1151 \\
f^{(1)}(x) &= 6x^2 - 8x \\
f^{(2)}(x) &= 12x - 8 \\
f^{(3)}(x) &= 12 \\
f^{(4)}(x) &= 0
\end{aligned}
$$

Let's find the first four derivatives of g.

$$
\begin{aligned}
g^{(0)}(x) &= x^3 \sin x \\
g^{(1)}(x) &= 3x^2 \sin x + x^3 \cos x \\
g^{(2)}(x) &= 6x \sin x + 3x^2 \cos x + 3x^2 \cos x - x^3 \sin x \\
&= (6x - x^3) \sin x + (6x^2) \cos x \\
g^{(3)}(x) &= (6 - 3x^2) \sin x + (6x - x^3) \cos x + (12x) \cos x - (6x^2) \sin x \\
&= (6 - 3x^2 - 6x^2) \sin x + (6x - x^3 + 12x) \cos x \\
&= (6 - 9x^2) \sin x + (18x - x^3) \cos x \\
g^{(4)}(x) &= (-18x) \sin x + (6 - 9x^2) \cos x + (18 - 3x^2) \cos x - (18x - x^3) \sin x \\
&= (-18x - 18x + x^3) \sin x + (6 - 9x^2 + 18 - 3x^2) \cos x \\
&= (-36x + x^3) \sin x + (24 - 12x^2) \cos x
\end{aligned}
$$